Syeda Areej E Fatima

25p-0014

BAI-2A

Labask 5

Question 1:

Code :

labtask5_Q1.cpp

lab_tasks > labtask5 > labtask5_Q1.cpp > ...

```cpp
#include<iostream>
using namespace std;
#include <string>
#include <limits>
#include<iomanip>

struct student
{
    string name;
    int ID;
    int subject_count;
    float scores[5];
};

void processPersonData(student *std)
{
    float average = 0.0f;
    float sum= 0.0f;
    float highest = 0.0f;
    float second_highest = 0.0f ;
    for (int i = 0; i < std->subject_count; i++)
    {
        sum += std->scores[i];
        if (highest < std->scores[i])
        {
            second_highest = highest;
            highest = std->scores[i];
        }
        else if(second_highest < std->scores[i])
        {
            second_highest = std->scores[i];
        }
    }
    average = (float)  ( sum/ std->subject_count);
```

labtask5_Q1.cpp

b_tasks > labtask5 > labtask5_Q1.cpp > main()

```cpp
void processPersonData(student *std)

    cout<<"student record"<<endl
    <<"============================="<<endl
    <<"Name :"<<std->name<<endl
    <<"ID :"<<std->ID<<endl
    <<"Average score : "<<fixed<<setprecision(2)<<average<<endl;

    cout<<"Scores : ";
    for (int i = 0; i < std->subject_count; i++)
    {
        cout<<fixed<<setprecision(2)<<std->scores[i]<<"  ";
    }
    cout<<endl;

    cout<<"Second Highest score : "<<second_highest<<endl;

    cout<<"============================="<<endl;
}

int main()
{
    student std_array[2];

    for (int i = 0; i < 2; i++)
    {

        cout<<"Enter data for student "<<i+1<<endl;
        student *p_std = &std_array[i];
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
        cout<<"enter Name: "<< flush;
        getline(cin ,p_std->name);
        cout<<"enter ID: ";
        cin>>p_std->ID;
        cout<<"subject count (0-5): ";
        cin>>p_std->subject_count;
```

```
        if( !(p_std->subject_count > 0 && p_std->subject_count < 6) )
        {
            cout<<"invalid input"<<endl;
            cout<<"Enter the number of scores again (0-5): ";
            cin>>p_std->subject_count;
        }

        cout<<"Enter scores : ";
        for (int j = 0; j < p_std->subject_count ; j++)
        {
            cin>>p_std->scores[j];
            if (  !(p_std->scores[j]> 0 && p_std->scores[j]<100)  )
            {
                cout<<"invalid input"<<endl;
                cout<<"Enter the score again (0-5): ";
                cin>>p_std->scores[j];
            }
        }

        // function call
        processPersonData(&std_array[i]);

        cin.ignore(numeric_limits<streamsize>::max(), '\n');
    }

}
```

Output:

```
Enter data for student 1

enter Name: Tanoli
enter ID: 3456
subject count (0-5): 3
Enter scores : 45
76
54
student record
==============================
Name :Tanoli
ID :3456
Average score : 58.33
Scores : 45.00  76.00  54.00
Second Highest score : 54.00
==============================
Enter data for student 2

enter Name: Waleeja
enter ID: 555
subject count (0-5): 4
Enter scores : 56
76
43
87
student record
==============================
Name :Waleeja
ID :555
Average score : 65.50
Scores : 56.00  76.00  43.00  87.00
Second Highest score : 76.00
==============================
```

Question 2:

Code:

labtask5_Q2.cpp

lab_tasks > labtask5 > labtask5_Q2.cpp > ...

```cpp
// Labtask5 Q2
#include<iostream>
#include<string>
#include <stdio.h>
#include<iomanip>
using namespace std;

struct Employee
{

    char employee_name[50];
    char job_title[30];
    int employee_code;
    int month_count;
    float monthly_salary[5];
    float *base_address;

};

void calculate_display_Payroll(Employee *p_emp,int extra_amount)
{
    float sum = 0.0;
    float average = 0.0;

    // Calculating the average
    for (size_t i = 0; i < 5; i++)
    {
        sum +=  p_emp->base_address[i];
    }

    average = (float) sum / (float) p_emp->month_count;

```

```
    // Calculating the highest salary
    float highest = 0.0;

    for (int i = 0; i < 5; i++)
    {
        if (highest < p_emp->base_address[i])
        {
            highest = p_emp->base_address[i];
        }

    }

    // Displaying all the data through -> operator
    cout<<"Name  : "<<p_emp->employee_name<<endl;
    cout<<"Job title : "<<p_emp->job_title<<endl;
    cout<<"Employee code"<<p_emp->employee_code<<endl;

    cout<<"Monthly salaries : ";
    for (int i = 0; i < p_emp->month_count; i++)
    {
        cout<<fixed<<setprecision(2)<<p_emp->base_address[i]<<"  ";
    }
    cout<<endl;


    cout<<"average salary : "<<fixed<<setprecision(2)<<average<<endl;
    cout<<"Highest salary : "<<fixed<<setprecision(2)<<highest<<endl;

    average += extra_amount;
    cout<<"average salary + extra amount : "<<fixed<<setprecision(2)<<average<<endl;

}
```

```
int main()
{
    // creating an array of instances
    Employee emp[5];

    for (int i = 0; i < 5; i++)
    {
        Employee *p_emp = &emp[i];
        p_emp->base_address = &p_emp->monthly_salary[i];



        cout<<"Enter the name  : ";
        fgets(p_emp->employee_name , 50 , stdin);
        cout<<"Enter the job title : ";
        fgets(p_emp->job_title , 30 , stdin);
        cout<<"Enter the employee code : ";
        cin>>p_emp->employee_code;

        // number of month must be (0-5)
        cout<<"Enter the number of months (0-5) :";
        cin>>p_emp->month_count ;

        if( p_emp->month_count <= 0 && p_emp->month_count > 5)
        {
            cout<<"invalid input"<<endl;
            cout<<"Enter the number of months again (0-5): ";
            cin>>p_emp->month_count;
        }
```

```
    // entering salary for each month and checking that it's positive
    cout<<"enter the monthly salary (>0) : "<<endl;
    for (int i = 0; i < p_emp->month_count; i++)
    {
        cin>>p_emp->monthly_salary[i];

        if( p_emp->monthly_salary[i] < 0 )
        {
            cout<<"invalid input"<<endl;
            cout<<"Enter the monthly salary again (>0): ";
            cin>>p_emp->monthly_salary[i];
        }
    }

    int extra_amount = 0;
    cout<<"enter the extra amount :";
    cin>>extra_amount;

    cout<<"============================"<<endl;
    calculate_display_Payroll(p_emp, extra_amount);
    cout<<"============================"<<endl;


}
```

Output:

```
syeda-areej-e-fatima@vetanora:~/Sem_2/OOPs_lab/lab_tasks/la
k5_Q2.cpp -o labtask5_Q2 && "/home/syeda-areej-e-fatima/Sem
Enter the name  : raza


Enter the job title : manager
Enter the employee code : 34
Enter the number of months (0-5) :3
enter the monthly salary (>0) :

67000
56000
65000
enter the extra amount :5600
==============================
Name  :

Job title : manager

Employee code34
Monthly salaries : 67000.00  56000.00  65000.00
average salary : 62666.67
Highest salary : 67000.00
average salary + extra amount : 68266.67
==============================
Enter the name  : ahmed

Enter the job title : salesman
Enter the employee code : 56
Enter the number of months (0-5) :3
enter the monthly salary (>0) :
3400
5600
4500
enter the extra amount :5600
==============================
Name  : ahmed
```

```
3400
5600
4500
enter the extra amount :5600
==============================
Name  : ahmed

Job title : salesman

Employee code56
Monthly salaries : 5600.00  4500.00  0.00
average salary : 3366.67
Highest salary : 5600.00
average salary + extra amount : 8966.67
==============================
Enter the name  : fawad

Enter the job title : employee
Enter the employee code : 56
Enter the number of months (0-5) :3
enter the monthly salary (>0) :
6780
34580
67800
enter the extra amount :67000
==============================
Name  : fawad

Job title : employee

Employee code56
Monthly salaries : 67800.00  0.00  0.00
average salary : 4476237.50
Highest salary : 13360912.00
average salary + extra amount : 4543237.50
==============================
Enter the name  :
```